

Problem 2

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Problem 1 Sketch a possible graph of a function g that satisfies the following conditions:

$$\text{Domain: } [-5, -2) \cup (-2, 3) \cup (3, 5)$$

$$\lim_{x \rightarrow -2} g(x) = 3$$

$$g(1) = 1$$

$$\lim_{x \rightarrow 1^+} g(x) = -\infty$$

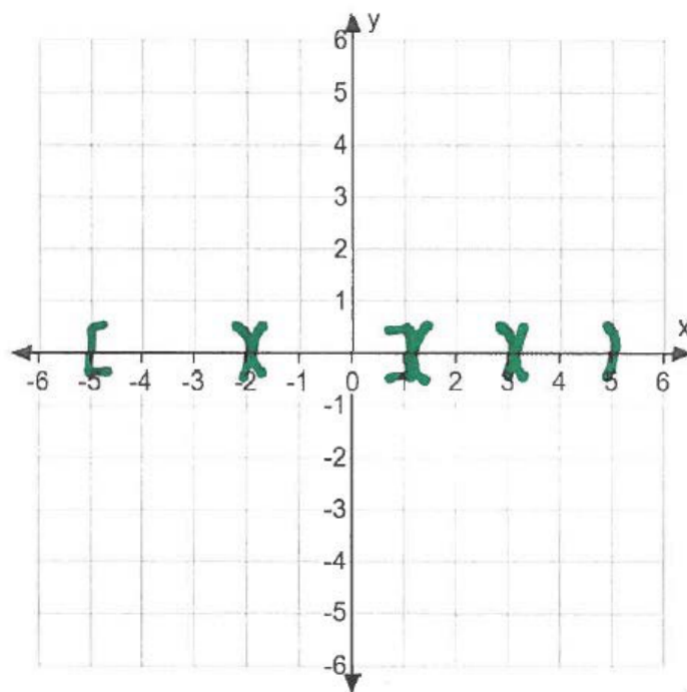
$$\lim_{x \rightarrow 3} g(x) = \infty$$

$$\lim_{x \rightarrow 5^-} g(x) = \infty$$

$$\lim_{x \rightarrow 1^-} g(x) = 1$$

$$g(-5) = -1.8$$

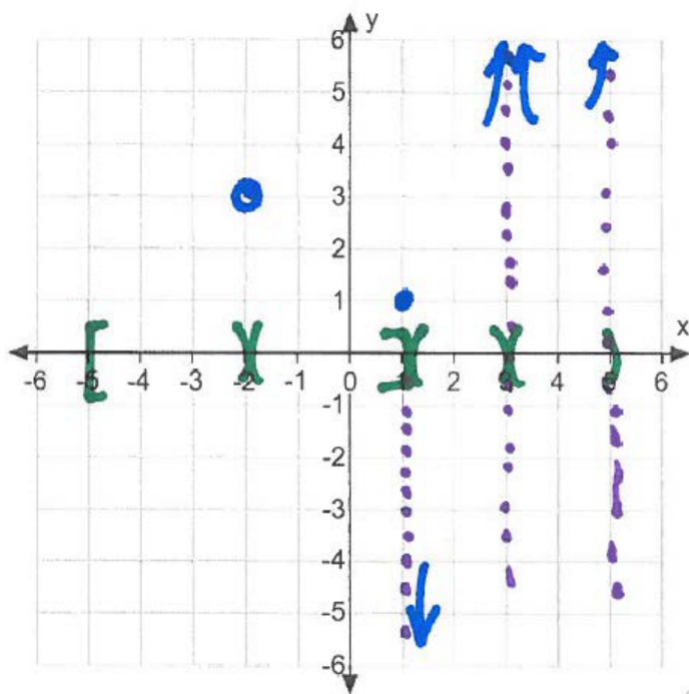
Explanation. First we will draw our domain on the x-axis using green. We will use a bracket when a point is in the domain and curved parenthesis when it is not.



Next, we draw an open circle at $(-2, 3)$ because we know $\lim_{x \rightarrow -2} g(x) = 3$. We also draw a closed circle at $(1, 1)$ and $(-5, -1.8)$ because we know $g(1) = 1$ and

Problem 2

$g(-5) = -1.8$. Finally we draw in our vertical asymptotes in purple as indicated by the infinite limits at $x = 1, x = 3, x = 5$. We draw the tails of the asymptotes in blue to indicate whether the function g approaches $-\infty$ or ∞ .



Finally, we connect our graph together.

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